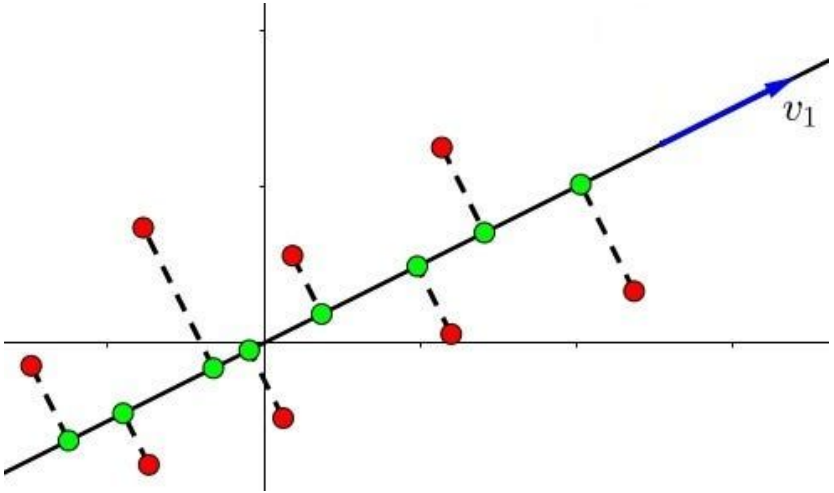
# Dimensionality reduction

Nonlinear methods

# Linear transformations - last time



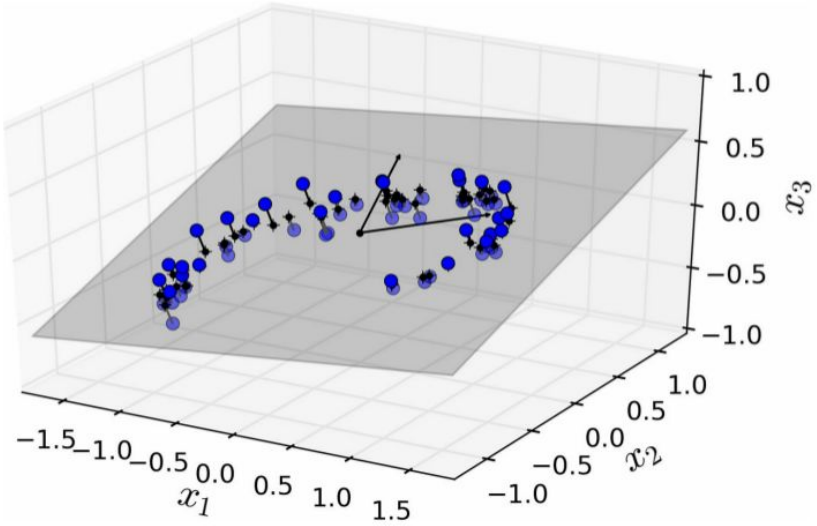



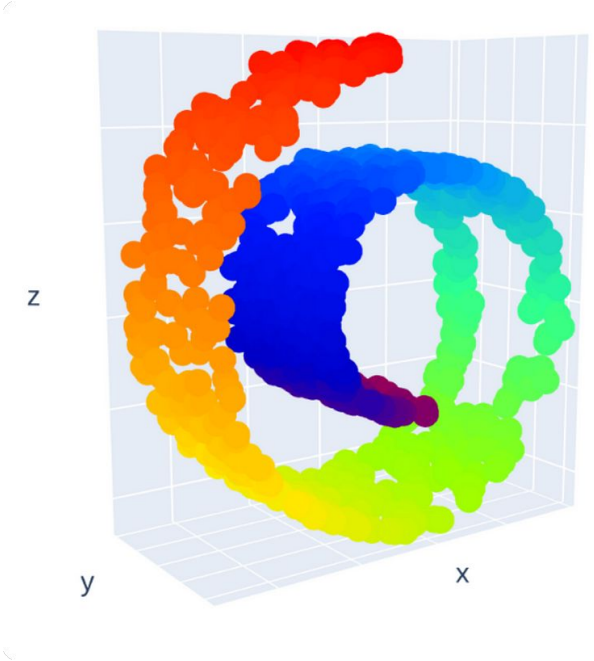

*Figure 8-2. A 3D dataset lying close to a 2D subspace*

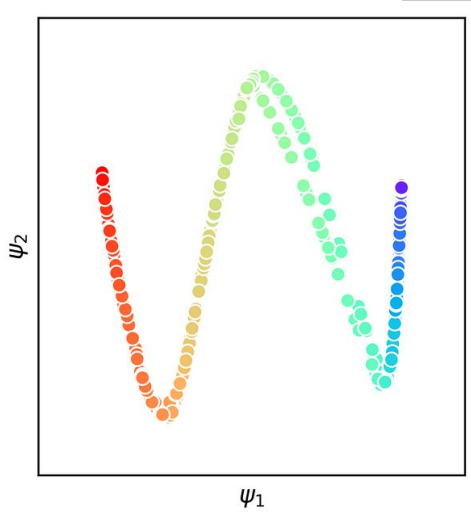
# Linear transformations - last time

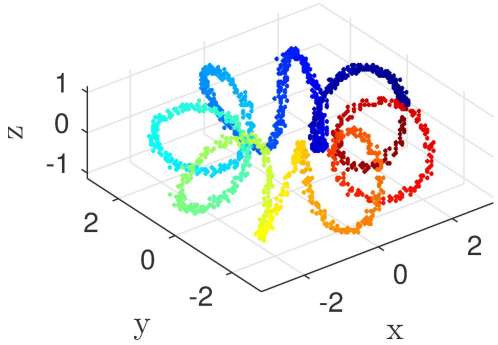
$$e(\mathcal{W}) = \sum_{i=1}^{n} \|\mathbf{a}_i - \mathbf{P}_{\mathcal{W}}\mathbf{a}_i\|_2^2 = \sum_{i=1}^{n} \|\mathbf{a}_i - WW^T\mathbf{a}_i\|_2^2 = \|A - WW^TA\|_F^2$$

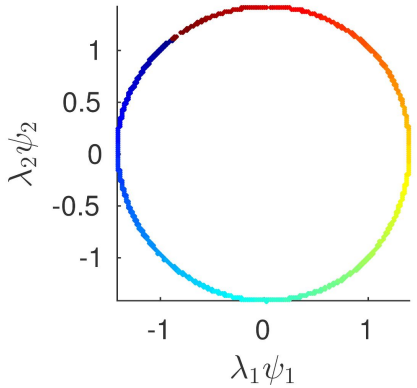
$$\|A - WW^TA\|_F \longrightarrow \min_{W, W^TW = I_k}$$

# Nonlinear transformations - today



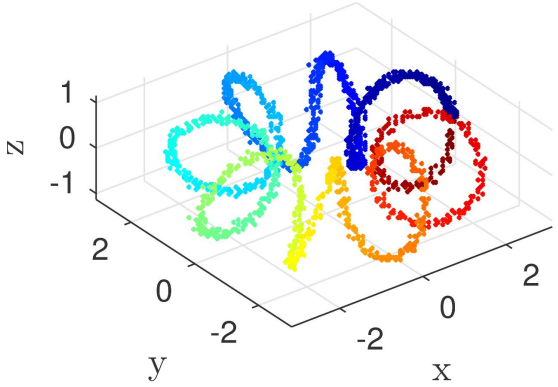
# Motivation

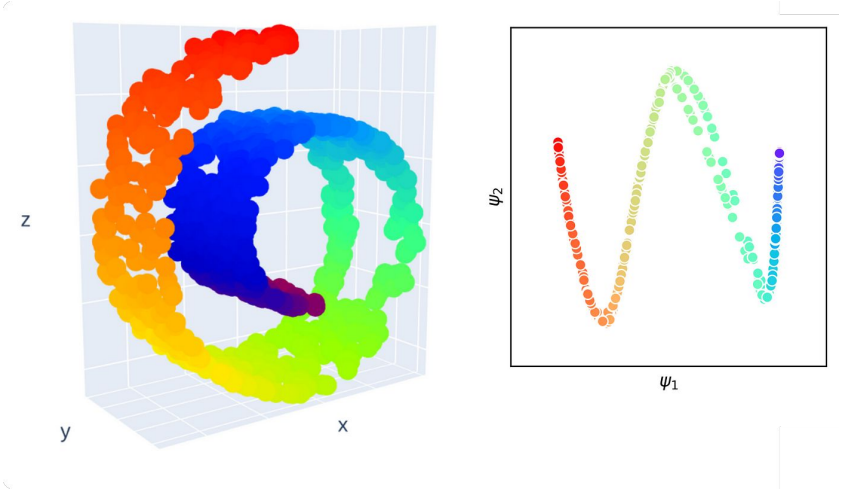
- last time: learning linear subspaces
- today: learning nonlinear **manifolds**

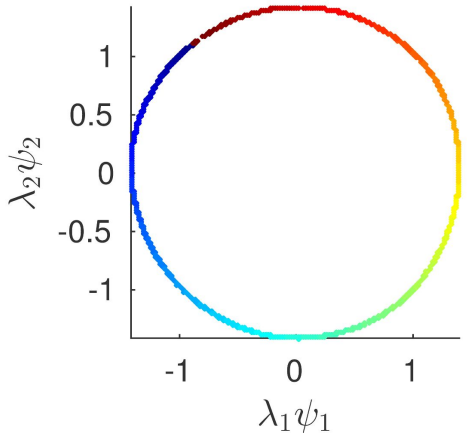
- possible applications:
  - using data assumed to lie on some lower-dimensional manifold, define new coordinates in which linear techniques can be used
  - the new coordinates should enable dimensionality reduction of the problem
  - the new coordinates should enable efficient clustering techniques

# Content

1. Diffusion maps
2. Autoencoders

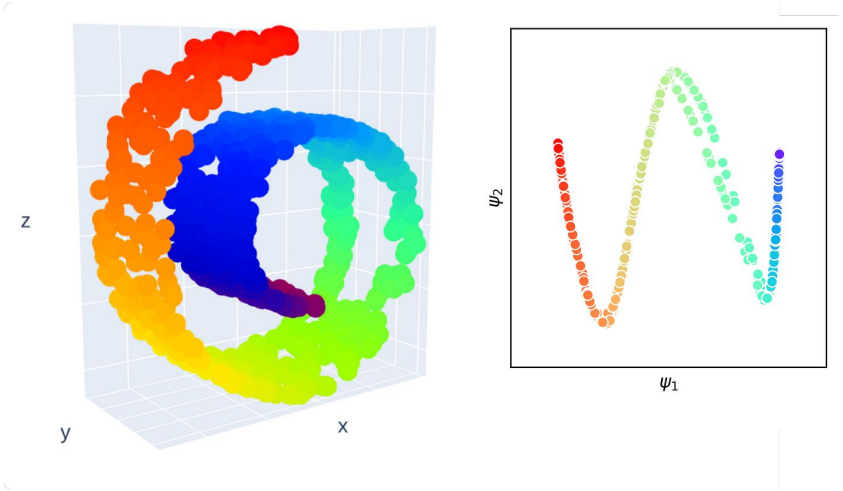


# Diffusion maps

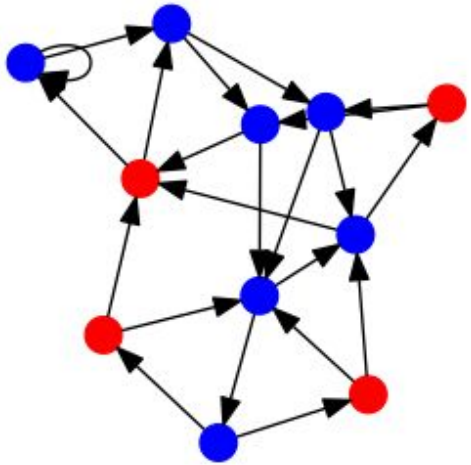
- we are given some set of points in high-dimensional space



# Diffusion maps

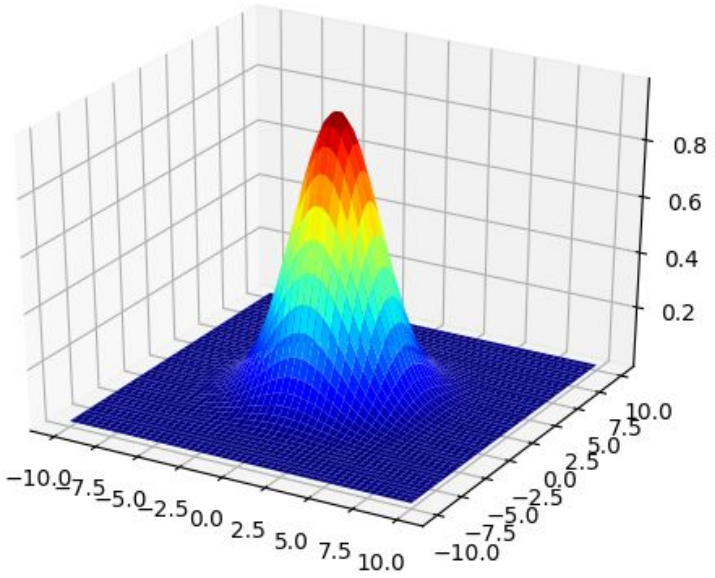
- random walk on a graph
- define probability *p*(x, y) of jump from node x to node y
- near nodes have greater probability
- based on, for example, Gaussian kernel:

$$k(x, y) = e^{-\|x-y\|_2^2/\sigma^2}$$



# Gaussian Kernel

$$k(x, y) = e^{-\|x-y\|_2^2/\sigma^2}$$



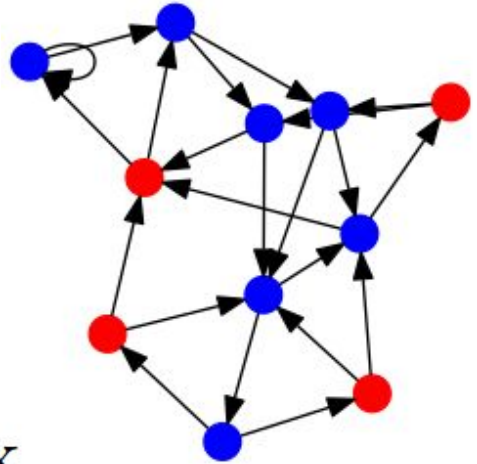
# Matrix formulation

$$K_{ij} = k(\mathbf{a}_i, \mathbf{a}_j)$$

$$d_i = \sum_j K_{ij}$$

$$D = diag(d_i)_{i=1}^{n}$$

$$P = D^{-1}K \text{ - row stochastic matrix}$$



P - Markov chain transition matrix

# Properties of the transition matrix P

$P = D^{-1}K = D^{-1/2}(D^{-1/2}KD^{-1/2})D^{1/2}$ is similar to matrix $P_s = D^{-1/2}KD^{-1/2}$

$P_s$ is symmetric, so it is also diagonalizable.

$P_s = U\Lambda U^T$, $\Lambda$ diagonal, $U$ orthogonal

$P = D^{-1/2}P_sD^{1/2} = D^{-1/2}U\Lambda U^TD^{1/2} = V\Lambda V^{-1}$

# Diffusion distance

$P^t$ probabilities? $P^t$ number of paths?

$$D_t(\mathbf{a}_i, \mathbf{a}_j)^2 = \sum_{x \in \{a_1, \ldots a_n\}} (P_t(a_i, x) - P_t(a_j, x))^2 \omega(x)$$

$t$ - number of steps, $\omega(a_k) = 1/d_k$

for $\Gamma_i = P^t(i, :)^T$ we have $D_t(\mathbf{a}_i, \mathbf{a}_j)^2 = (\Gamma_i - \Gamma_j)^T D^{-1} (\Gamma_i - \Gamma_j) = \|\Gamma_i - \Gamma_j\|^2_{D^{-1}}$

# Diffusion map

$$P = V\Lambda V^{-1},\ \ P^t = V\Lambda^t V^{-1},\ \ V = D^{-1/2}U,\ U^TU = I$$

$$\widetilde{V} = V^{-1} \Rightarrow \widetilde{V}P = \Lambda\widetilde{V},\ \ \widetilde{V}P^t = \Lambda^t\widetilde{V}$$

$$P^t = V\Lambda^t\widetilde{V} \Rightarrow P^t(i,:) = \begin{pmatrix} v_1(i)\lambda_1^t & v_2(i)\lambda_2^t & \dots & v_n(i)\lambda_n^t \end{pmatrix}\widetilde{V}$$

$$\Gamma_i = (P^t(i,:))^T = \widetilde{V}^T \underbrace{\begin{pmatrix} \lambda_1^t v_1(i) \\ \lambda_2^t v_2(i) \\ \vdots \\ \lambda_n^t v_n(i) \end{pmatrix}}_{\Psi_i^{(t)}} = \widetilde{V}^T\Psi_i^{(t)}$$

$$D_t(\mathbf{a}_i, \mathbf{a}_j)^2 = (\Psi_i^{(t)} - \Psi_j^{(t)})^T \underbrace{\widetilde{V}D^{-1}\widetilde{V}^T}_{U^TD^{1/2}D^{-1}D^{1/2}U} (\Psi_i^{(t)} - \Psi_j^{(t)}) = \|\Psi_i^{(t)} - \Psi_j^{(t)}\|_2^2$$

# Diffusion map

$$\mathbf{a}_i \mapsto \Psi_i^{(t)} = \begin{pmatrix} \lambda_1^t v_1(i) \\ \lambda_2^t v_2(i) \\ \vdots \\ \lambda_n^t v_n(i) \end{pmatrix} = \Lambda^t V(i,:)^T$$ with parameter *t*

$$|\lambda_1| > |\lambda_2| \geq \cdots \geq |\lambda_n|, \quad |\lambda_1^t| \gg |\lambda_2^t| \geq \cdots \geq |\lambda_n^t|$$

Dimensionality reduction:

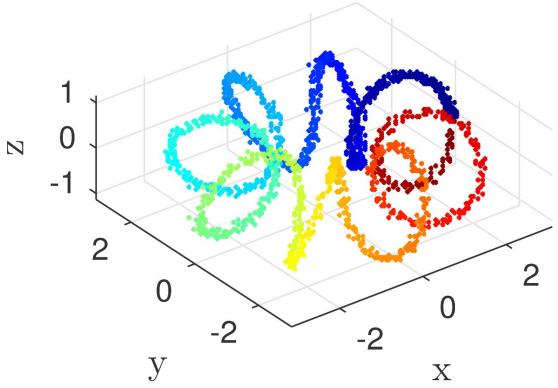
$$\Psi^{(t)} = \begin{pmatrix} \Psi_1^{(t)} & \cdots & \Psi_n^{(t)} \end{pmatrix} = \begin{pmatrix} x & x & x & x & x \\ x & x & x & x & x \\ x & x & x & x & x \\ x & x & x & x & x \end{pmatrix}$$

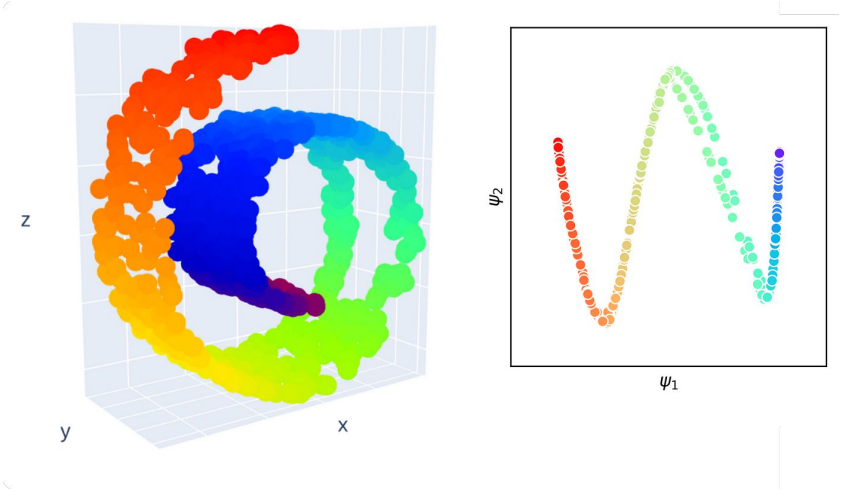
# Diffusion maps - the algorithm

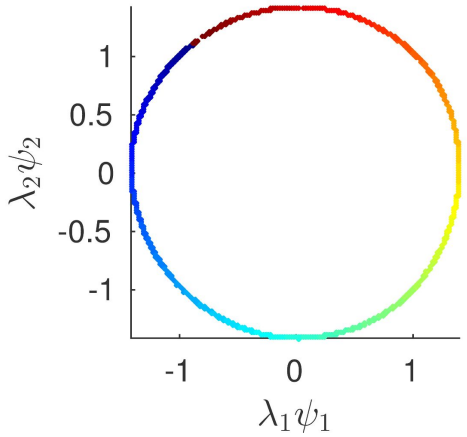
1. calculate matrix $K$ and $d = Ke$, where $e = (1, 1, \ldots, 1)^T$
2. calculate $P_s = D^{-1/2} K D^{-1/2}$ and its eigendecomposition $P_S = U \Lambda U^T$
3. calculate $V = D^{-1/2} U$
4. $\Psi^{(t)} = \Lambda^t V^T$
5. take first $k$ rows: $\Psi_k^{(t)} = \Psi^{(t)}(1 : k, 1 : n)$

# Diffusion maps - interesting use cases

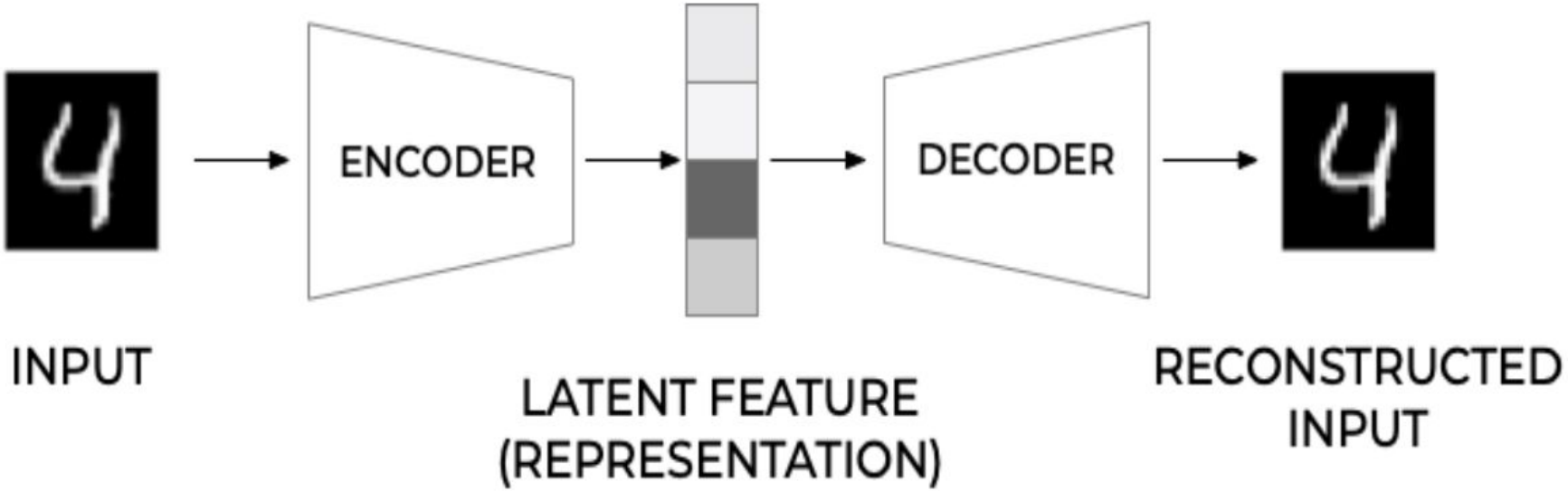
1. clustering with reduced dimensions
2. determine the correct order in time for a series of data



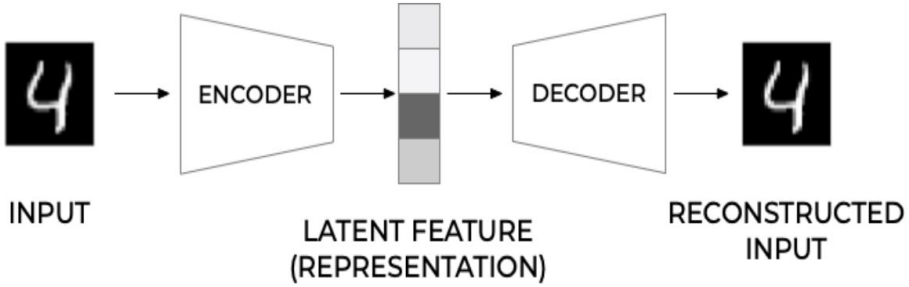



# Autoencoders



# Autoencoders




$$\mathbf{h}_i = g(\mathbf{x}_i)$$

$$\tilde{\mathbf{x}}_i = f\left(\mathbf{h}_i\right) = f\left(g\left(\mathbf{x}_i\right)\right)$$

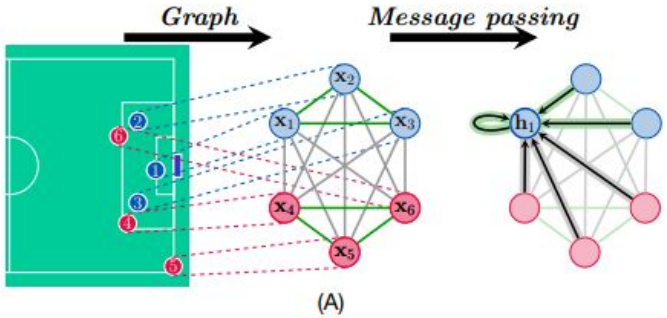
$$\arg\min_{f,g} < [\Delta(\mathbf{x}_i, f(g\left(\mathbf{x}_i\right))] >$$

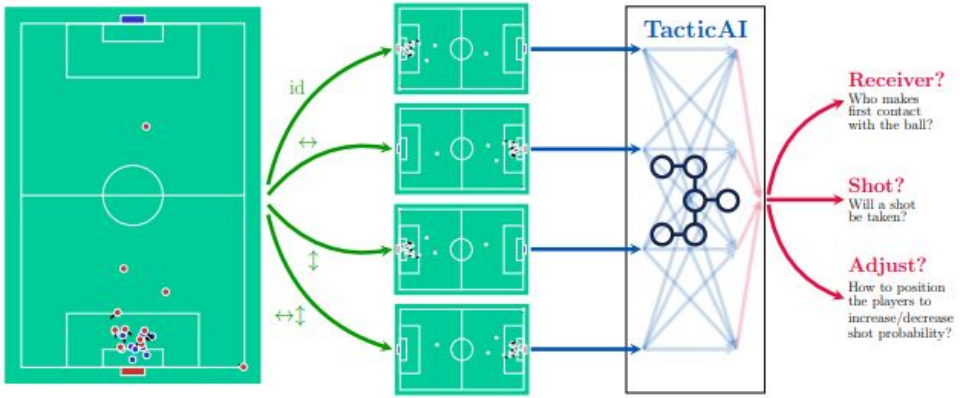
# Autoencoders

Relation to:

- PCA?
  - if autoencoder does not have any nonlinearities, it can, in the best but not very probable case, give us the same result as PCA
- Linear best-fit subspaces?
  - same, since PCA and best-fit subspaces are the same thing
- Diffusion maps?
  - both are non-linear

# Interesting application of (variational) autoencoders



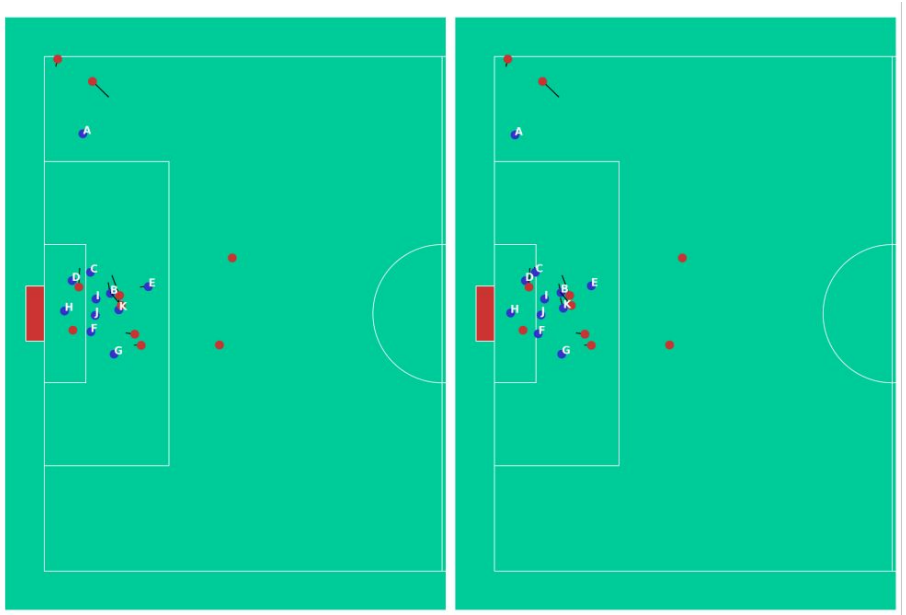




TacticAI paper





# Interesting application of (variational) autoencoders



Supplementary Figure 5 | **Sample of Task 4 in the case study.** In this task, the human raters are given pairs of corner kick samples, each of which consists of a reference corner kick sample and its adjustment generated by TacticAI. Then, the raters are asked to identify whether there are salient improvements in the adjustment and also to give specific account to those improvements as well. For the adjustment in each pair, we rate it as +1 if the rater confirms that overall adjustments are constructive, 0 if no distinct difference, or −1 if destructive.

# Summary

- We covered:
  - Diffusion maps
  - Autoencoders
    - TacticAI application

- Other popular nonlinear dimensionality reduction methods:
  - Isomap, locally linear embedding, Laplacian eigenmaps, t-SNE, UMAP
  - variational autoencoders

# References


R. R. Coifman, S. Lafon: Diffusion maps. Appl. Comput. Harmonic Analysis 21 (2006) 5-30.

Z. Drmac, MMP lectures, 2021.

Michelucci, Umberto. "An introduction to autoencoders." arXiv preprint arXiv:2201.03898 (2022).

Wang, Zhe, Petar Veličković, Daniel Hennes, Nenad Tomašev, Laurel Prince, Michael Kaisers, Yoram Bachrach et al. "TacticAI: an AI assistant for football tactics." *Nature communications* 15, no. 1 (2024): 1906.